

**SIMATS ENGINEERING
ASSESSMENT TOOL 3**

COURSE NAME: COMPUTER NETWORKS
COURSE CODE: CSA0713

COURSE OUTCOME COVERED

CO2: Configure IP addressing schemes, subnetting, and basic routing techniques using simulation tool, for efficient network design and topology. (BL3)

Assessment Tool Weightage: 30%

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Annexure A

Debugging & Optimisation Task

Code of Conduct:

*I S.Karthikeyan(192419048) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student:

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Annexure A

Debugging & Optimisation Task

Question 1

Debugging and Optimization of WLAN and WWAN Infrastructure

Introduction

A Wireless Local Area Network (WLAN) enables employees within an organization to connect laptops, smartphones, printers, IP phones, and IoT devices through Wi-Fi. A Wireless Wide Area Network (WWAN) allows employees working outside the office to access company servers and applications using 4G/5G mobile networks. In the given scenario, the IT company operates from a six-storey smart office with more than 1,200 employees who rely on wireless communication for cloud computing, online meetings, file sharing, and video conferencing. Recently, the company has been facing several connectivity and performance issues affecting daily operations. As a network engineer, it is important to identify the root causes of these problems and recommend suitable debugging and optimization techniques.

Problems Identified

The following problems were identified in the existing wireless network:

- Slow internet speed during peak working hours.
 - Frequent Wi-Fi disconnections while employees move between floors.
 - Weak wireless signals in conference rooms.
 - Video conferencing applications freeze or disconnect.
 - Some wireless access points are overloaded while others remain underutilized.
 - Field engineers experience poor WWAN connectivity in rural areas.
 - Some wireless devices still use outdated security protocols.
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Root Cause Analysis

The main reasons for these problems are:

1. **Network Congestion:** Too many users are connected to a single access point, causing bandwidth congestion and slow internet performance.

2. **Poor Access Point Placement:** Access points are not installed in suitable locations, resulting in weak signal strength and dead zones.
 3. **Improper Roaming Configuration:** Devices cannot switch smoothly between access points while moving across floors, leading to frequent Wi-Fi disconnections.
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Debugging Techniques

The following debugging steps should be performed to identify and resolve the problems:

- Monitor network traffic using tools such as Wireshark, Cisco DNA Center, or PRTG.
 - Perform a wireless site survey to detect weak signal areas and interference.
 - Check access point configuration, channel allocation, and transmit power.
 - Test Wi-Fi roaming between different floors.
 - Measure WWAN signal strength and latency in different locations.
 - Conduct a wireless security audit and update firmware on all networking devices.
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Optimization Techniques

To improve the performance and reliability of the wireless network, the following optimization techniques should be implemented:

1. Deploy Additional Access Points

Install additional access points in crowded areas to reduce congestion and improve coverage.

2. Enable Load Balancing

Distribute users evenly among available access points to prevent overloading and improve network performance.

3. Upgrade to Wi-Fi 6

Implement IEEE 802.11ax (Wi-Fi 6) to support more users with higher speed, lower latency, and better efficiency.

Benefits of Optimization

After implementing the above solutions, the organization will achieve:

- Higher wireless speed and better throughput.
- Stronger Wi-Fi signal across all floors.
- Seamless roaming without disconnections.
- Better video conferencing quality.

Conclusion

The company's wireless networking issues are mainly caused by congestion, poor access point placement, weak signal coverage, improper roaming, limited WWAN coverage, and outdated security protocols. By following a systematic debugging process and implementing optimization techniques such as additional access points, load balancing, Wi-Fi 6, fast roaming, QoS, improved WWAN coverage, and WPA3 security, the organization can significantly improve network performance, reliability, and security. These measures ensure uninterrupted communication, efficient resource utilization, and seamless connectivity for both office employees and remote field engineers.

Question 2

Debugging and Optimization of ATM (Asynchronous Transfer Mode) Network

Introduction

Asynchronous Transfer Mode (ATM) is a high-speed, connection-oriented networking technology that transfers data using fixed-size **53-byte cells** (5-byte header and 48-byte payload). It supports voice, video, and data transmission with high reliability and low delay. In the given scenario, the organization is experiencing ATM network issues such as high delay, packet loss, congestion, poor Quality of Service (QoS), and connection failures. These problems reduce network efficiency and affect real-time applications. Therefore, proper debugging and optimization techniques are required to improve the ATM network performance.

Problems Identified

The ATM network faces the following issues:

- High transmission delay during peak traffic.
- Packet (cell) loss due to network congestion.
- Poor Quality of Service (QoS) for voice and video applications.
- Frequent connection failures between ATM switches.
- Low bandwidth utilization.
- Increased response time for network applications.

These issues decrease the overall performance and reliability of the ATM network.

Root Cause Analysis

The major causes of these problems are:

1. **Network Congestion:** Too many ATM cells are transmitted simultaneously, causing buffer overflow and cell loss.
 2. **Improper Traffic Management:** Lack of effective traffic shaping and policing results in uneven traffic flow.
 3. **Faulty ATM Switches:** Hardware failures or incorrect switch configurations interrupt communication between network devices.
 4. **Insufficient Bandwidth:** The available bandwidth is not enough to handle increasing network traffic.
 5. **QoS Misconfiguration:** Real-time applications such as voice and video are not given higher priority, leading to delays and poor performance.
 6. **Connection Setup Errors:** Incorrect Virtual Path Identifier (VPI) and Virtual Channel Identifier (VCI) configurations can cause connection failures.
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Debugging Techniques

The following steps can be used to identify and resolve ATM network problems:

- Monitor ATM traffic using network monitoring tools.
- Check ATM switch status and hardware health.
- Verify VPI and VCI configurations.
- Analyze network congestion and cell loss statistics.
- Test bandwidth utilization and transmission delay.
- Review QoS settings for voice, video, and data traffic.
- Update firmware and replace faulty networking equipment if required.

These debugging techniques help identify the exact cause of performance degradation.

Optimization Techniques

1. Traffic Shaping

Control the rate of data transmission to reduce congestion and improve network stability.

2. Congestion Control

Implement congestion management techniques to minimize cell loss and improve network efficiency.

3. Quality of Service (QoS)

Prioritize delay-sensitive applications such as voice and video over normal data traffic.

4. Upgrade Network Bandwidth

Increase available bandwidth to support more users and high-speed applications.

5. Optimize ATM Switch Configuration

Configure ATM switches correctly by verifying VPI/VCI values and routing paths to ensure reliable communication.

6. Preventive Maintenance

Regularly inspect ATM devices, update firmware, and replace faulty hardware to maintain network reliability.

7. Continuous Network Monitoring

Use monitoring software to detect congestion, delays, and errors before they affect network performance.

Benefits of Optimization

After implementing these optimization techniques, the organization will achieve:

- Reduced network congestion.
- Lower transmission delay.
- Improved Quality of Service (QoS).
- Better voice and video communication.
- Increased bandwidth utilization.
- Reliable ATM connections.
- Improved network stability and performance.
- Higher customer and user satisfaction.

Conclusion

The ATM network problems are mainly caused by congestion, insufficient bandwidth, incorrect switch configuration, and poor Quality of Service management. By performing systematic debugging and implementing optimization techniques such as traffic shaping, congestion control, bandwidth upgrades, proper ATM switch configuration, QoS implementation, and continuous monitoring, the organization can significantly improve network performance, reduce delays, and ensure reliable communication for voice, video, and data services.

Question 3

Debugging and Optimization of MPLS (Multiprotocol Label Switching) Network

Introduction

Multiprotocol Label Switching (MPLS) is a high-performance networking technology that forwards data packets using **labels** instead of IP addresses. It provides faster packet forwarding, better Quality of Service (QoS), traffic engineering, and reliable communication between branch offices. In the given scenario, the organization is facing MPLS network issues such as routing failures, high latency, packet loss, traffic congestion, and poor application performance. These problems affect communication between different office locations and reduce overall network efficiency. Therefore, debugging and optimization techniques are required to improve the MPLS network.

Problems Identified

The following issues are observed in the MPLS network:

- High latency during data transmission.
- Packet loss between branch offices.
- Incorrect routing of data packets.
- Network congestion during peak hours.
- Poor performance of business applications.
- Failure of Label Switched Paths (LSPs).

These problems reduce the speed, reliability, and efficiency of the network.

Root Cause Analysis

The main causes of these issues are:

1. **Incorrect Label Distribution:** Improper label assignment causes routing errors and communication failures.
 2. **Network Congestion:** Heavy traffic leads to increased delay and packet loss.
 3. **Faulty Router Configuration:** Incorrect MPLS or routing protocol settings affect data forwarding.
 4. **LSP Failure:** Failure of Label Switched Paths interrupts communication between source and destination.
 5. **Insufficient Bandwidth:** Limited bandwidth cannot support increasing network traffic.
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Debugging Techniques

To identify and resolve MPLS issues, the following steps should be performed:

- Monitor network traffic and router performance.
- Verify MPLS label distribution and forwarding tables.
- Check Label Switched Path (LSP) status.
- Test routing protocols and router configurations.
- Measure network latency, packet loss, and bandwidth utilization.
- Analyze QoS settings for different types of traffic.
- Update router firmware and replace faulty devices if necessary.

These steps help locate the source of network failures and improve troubleshooting.

Optimization Techniques

1. Optimize Label Distribution

Ensure labels are correctly assigned and updated to avoid routing errors.

2. Traffic Engineering

Use MPLS Traffic Engineering (MPLS-TE) to distribute traffic efficiently and avoid network congestion.

3. Improve Quality of Service (QoS)

Prioritize important applications such as voice, video conferencing, and business-critical services.

4. Upgrade Bandwidth

Increase network bandwidth to support growing traffic and improve application performance.

5. Maintain Reliable LSPs

Monitor and maintain Label Switched Paths to ensure uninterrupted communication between branch offices.

6. Regular Network Monitoring

Continuously monitor router performance, traffic load, and network health to detect issues early.

7. Update Router Configuration

Regularly review and update router configurations to improve performance and maintain compatibility.

Benefits of Optimization

After implementing these solutions, the organization will achieve:

- Faster packet forwarding.
 - Reduced latency and packet loss.
 - Improved Quality of Service (QoS).
 - Better utilization of network bandwidth.
 - Reliable communication between branch offices.
 - Enhanced performance of business applications.
 - Increased network reliability and efficiency.
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Conclusion

The MPLS network problems are mainly caused by incorrect label distribution, network congestion, faulty router configuration, LSP failures, and insufficient bandwidth. By performing systematic debugging and implementing optimization techniques such as traffic engineering, QoS improvement, bandwidth upgrades, regular monitoring, and proper router configuration, the organization can significantly improve network performance, reliability, and scalability. These measures ensure fast, secure, and efficient communication across all branch offices.

Question 4

Debugging and Optimization of Error Detection and Error Correction Mechanisms

Introduction

Error detection and error correction are essential techniques in computer networks to ensure reliable data transmission. During communication, data may become corrupted due to noise, interference, or hardware failures. Error detection techniques identify whether an error has occurred, while error correction techniques detect and correct errors without retransmitting the data. In the given scenario, the network experiences frequent transmission errors, corrupted data, and reduced communication reliability. Proper debugging and optimization are required to improve data integrity and network performance.

Problems Identified

The following issues are observed in the network:

- Frequent transmission errors.
- Corrupted data packets during communication.
- Increased packet retransmissions.
- Reduced network reliability.
- Poor performance during data transfer.
- Delay in communication due to repeated error checking.

These problems affect the efficiency and accuracy of data communication.

Root Cause Analysis

The major causes of these problems are:

1. **Noise and Signal Interference:** Electrical noise and wireless interference corrupt transmitted data.
 2. **Damaged Communication Media:** Faulty cables or weak wireless signals increase the probability of transmission errors.
 3. **High Network Congestion:** Heavy traffic leads to packet loss and data corruption.
 4. **Improper Error Detection Method:** Weak error detection mechanisms fail to identify all transmission errors.
 5. **Hardware Failures:** Faulty network devices can introduce errors during data transmission.
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Debugging Techniques

To identify and resolve transmission errors, the following steps should be performed:

- Monitor the network for packet loss and error rates.
- Check the condition of communication cables and wireless links.
- Test network devices such as switches and routers.
- Verify the implementation of CRC, checksum, or Hamming code.
- Measure signal quality and interference levels.
- Analyze retransmission statistics using network monitoring tools.

These debugging techniques help identify the exact source of communication errors.

Optimization Techniques

1. Use Cyclic Redundancy Check (CRC)

Implement CRC to detect transmission errors with high accuracy before data reaches the receiver.

2. Apply Hamming Code

Use Hamming Code to detect and correct single-bit errors without requiring retransmission.

3. Improve Communication Media

Replace damaged cables and improve wireless signal strength to reduce transmission errors.

4. Reduce Network Congestion

Implement traffic management techniques to minimize packet loss and improve communication quality.

5. Regular Hardware Maintenance

Inspect and replace faulty routers, switches, and network interface cards to improve reliability.

6. Continuous Network Monitoring

Monitor error rates, packet loss, and network performance regularly to identify issues at an early stage.

Benefits of Optimization

After implementing these solutions, the organization will achieve:

- Accurate data transmission.
 - Reduced packet loss and retransmissions.
 - Improved network reliability.
 - Faster communication.
 - Better error detection and correction.
 - Increased overall network performance.
 - Higher user satisfaction.
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Conclusion

The network errors are mainly caused by noise, interference, damaged communication media, congestion, and hardware failures. By implementing effective error detection and correction techniques such as **CRC** and **Hamming Code**, along with proper network maintenance, congestion control, and continuous monitoring, the organization can significantly improve data accuracy, communication reliability, and overall network performance. These measures ensure secure and efficient transmission of information across the network.

Question 5

Debugging and Optimization of Flow Control Mechanisms

Introduction

Flow control is a technique used in computer networks to regulate the rate of data transmission between the sender and receiver. It ensures that the sender does not transmit data faster than the receiver can process it. Effective flow control prevents packet loss, buffer overflow, and network congestion. In the given scenario, the organization experiences slow data transfer, packet loss, buffer overflow, and reduced network performance due to inefficient flow control. Therefore, proper debugging and optimization techniques are required to improve communication efficiency.

Problems Identified

The following issues are observed in the network:

- Packet loss during data transmission.
 - Buffer overflow at the receiving device.
 - Slow network performance.
 - High transmission delay.
 - Reduced throughput.
 - Frequent retransmissions due to congestion.
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Root Cause Analysis

The major causes of these problems are:

1. **Sender Transmits Data Too Quickly:** The sender sends data faster than the receiver can process it, causing packet loss.
 2. **Buffer Overflow:** Limited buffer capacity at the receiver results in data loss.
 3. **Network Congestion:** Heavy traffic increases delays and reduces network efficiency.
 4. **Improper Flow Control Mechanism:** Inefficient implementation of flow control
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Debugging Techniques

The following steps should be performed to identify and resolve flow control issues:

- Monitor network traffic and throughput.
 - Check sender and receiver transmission rates.
 - Analyze buffer utilization and packet loss.
 - Measure network delay and congestion levels.
 - Verify flow control protocol configuration.
 - Inspect network devices for hardware or software faults.
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Optimization Techniques

1. Implement Sliding Window Protocol

Use the Sliding Window Protocol to allow efficient transmission of multiple frames while controlling the flow of data.

2. Use Stop-and-Wait ARQ Where Required

For simple and low-speed communication, Stop-and-Wait ARQ ensures reliable data transmission by waiting for an acknowledgment before sending the next frame.

3. Increase Buffer Capacity

Provide sufficient buffer memory at the receiver to avoid overflow during heavy traffic.

4. Control Network Congestion

Implement congestion control techniques to reduce packet loss and improve throughput.

5. Upgrade Network Bandwidth

Increase available bandwidth to support more users and improve transmission speed.

6. Continuous Network Monitoring

Regularly monitor traffic, throughput, and packet loss to detect problems before they affect network performance.

Benefits of Optimization

After implementing these solutions, the organization will achieve:

- Reduced packet loss.
 - Improved throughput.
 - Better utilization of network resources.
 - Faster and more reliable communication.
 - Reduced transmission delay.
 - Increased network efficiency.
 - Higher overall system performance.
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Conclusion

The flow control problems are mainly caused by high transmission rates, buffer overflow, congestion, and insufficient bandwidth. By implementing effective flow control mechanisms such as the **Sliding Window Protocol**, **Stop-and-Wait ARQ**, increased buffer capacity, congestion control, and continuous network monitoring, the organization can significantly improve data transmission, reduce packet loss, and enhance overall network performance. These measures ensure reliable, efficient, and stable communication across the network.